High-EQ Reply Assistant

A Generative AI Approach to Complex Social Context Response Generation

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The Pain Point: "Social Aphasia" in Modern Digital Communication



Core Conflict: High-pressure Social Scenarios

In today's digital world, we are constantly bombarded with **high-stakes social interactions**.

Whether it's politely declining an unreasonable request from a superior, navigating a tricky interperson conflict, or responding to an awkward personal question, the pressure to get the reply 'just right' is immense.

The consequences of a poorly worded message can be severe, potentially damaging relationships or professional opportunities.



User Pain Point: Mental Exhaustion & Communication Dilemma

This pressure often leads to a state we call **'social aphasia'** — a mental block where the fear of making a mistake paralyzes us, preventing us from forming a coherent and effective response.

This phenomenon is not just an inconvenience; it's a significant source of daily stress and anxiety.

Market Gap & Research Objectives



Market Status: Entertainment-focused, Functionally Lacking

While the market is flooded with AI applications designed for entertainment—like virtual pets or fantasy game companions—there is a striking absence of tools that solve real, pressing problems.

These 'nice-to-have' apps don't address the 'must-have' need for navigating social crises.

Research Objective: Building a "Social Survival Agent"

Our goal is to bridge this gap by creating a practical, AI-driven solution. We aim to leverage Large Language Models (LLMs) to create a **"social survival agent"** capable of generating customized, boundary-aware responses with varied tones.

The Breakthrough: Why GenAI is the Only Viable Tool



Farewell to Rule Trees: Limitations of Traditional Methods

The inherent complexity of human social interaction makes traditional programming methodologies largely ineffective.

Social exchanges are **infinitely variable and culturally situated**. Attempting to hard-code a rule-based conversational agent using conventional 'if-then' decision trees is an exercise in futility.

Faced with countless workplace scenarios, interpersonal frictions, and implicit social rules, **static rule trees simply cannot exhaust all possibilities**.



Core Advantage: Zero-Shot Style Transfer

Generative AI, particularly advanced LLMs, is the only viable technical solution because of its **intrinsic capacity for unstructured contextual understanding**. Their unique aptitude for zero-shot stylistic transfer allows us to dynamically recalibrate their output to embody desired personas through precise prompt engineering.

AI's "Theory of Mind"



Core Advantage: Simulating Recipient's Psychological Expectations

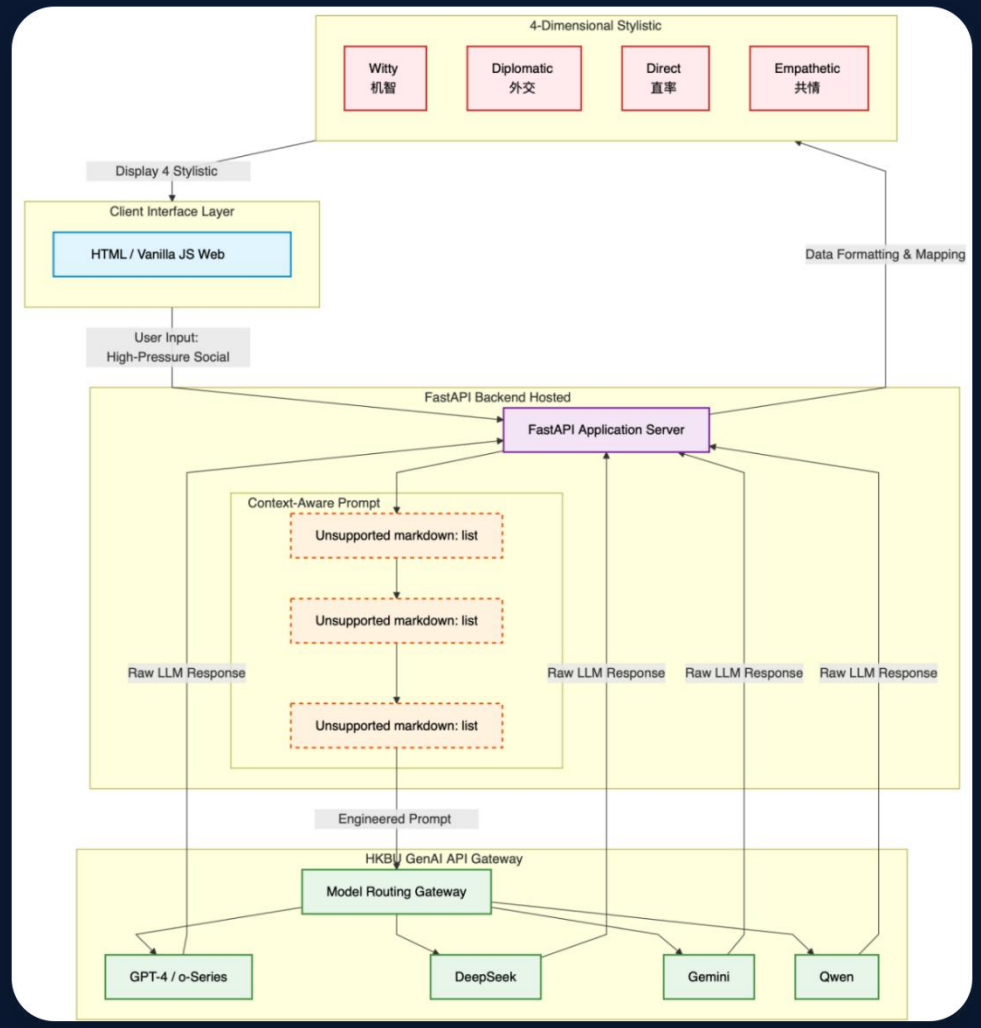
A key advantage of state-of-the-art LLMs is their demonstrated, albeit rudimentary, architectural capacity for **'Theory of Mind'**—the psychological ability to anticipate how a specific cognitive or emotional stance might be received by a counterpart.



Application: Internal "Multi-Agent Sandbox Simulation"

We leverage this capability by instructing the model to actively simulate the recipient's perspective. It effectively performs an internalized **'Multi-Agent Simulation' or sandbox negotiation**, evaluating potential conversational fallout before finalizing the output.

System Architecture: The "LLM EQ Arena"



Client Interface Layer

Lightweight, high-frequency interaction interface built with pure HTML/Vanilla JS.

FastAPI Backend

High-performance FastAPI framework deployed on the Railway cloud platform.

GenAI API Gateway

Integration with multiple state-of-the-art models (GPT/DeepSeek/Gemini/Qwen) via the HKBU GenAI API, enabling cross-validation and comparison.

Core Intervention Mechanism: The "System 2" Prompt Architecture

Break the limitations of AI's "intuitive response", force the model to implement a structured deep thinking protocol, and reshape the boundaries of interaction.

01



01 Implicit Intent Analysis

The model must first execute an intent analysis to detect any passive-aggressive or adversarial elements in the sender's text.

02



02 Behavioral Constraint Setting

It then applies strict behavioral constraints to suppress the AI's natural predisposition toward excessive compliance and unwarranted apologies.

03



03 Four-Dimensional Style Generation

Finally, it bifurcates its generative process to yield exactly four tonal variations: Witty, Diplomatic, Direct, and Empathetic, ensuring the user retains full decision-making power.

The Evaluation Benchmark: The Four-Dimensional EQ Evaluation Matrix



Dimension 1: Empathy & Politeness

Does the response validate the other party's feelings while maintaining dignity?



Dimension 2: Boundary Management

Does the response successfully defend the user's interests without over-conceding?



Dimension 3: Clarity & Effectiveness

Is the core message communicated efficiently?



Dimension 4: Context Appropriateness

Does the response respect the situational power dynamics (e.g., adjusting deference for peers vs. superiors)?

Implementation & Evolution Path (Ablation Study)



Control Group (Baseline)

We used the unconstrained, zero-shot native output of a standard LLM as our baseline for comparison.



Experimental Group

We introduced our Context-Aware Prompt Architecture and compared its output directly against the baseline in blind user tests.

Core Value of the Experiment: Quantifying Intervention Efficacy

This rigorous design allows us to isolate and measure the exact impact of our intervention on response quality.

Key Finding: Bridging the Gap Between Robotic and Passive-Aggressive

Phase 1: Too Robotic

ROBOTIC STYLE

The baseline model produced responses users deemed highly artificial and 'robotic', automatically apologizing or sacrificing the user's boundaries.

Phase 2: Overcompensation

OVER-EXPRESSIVE

Early attempts to inject personality caused the prompts to swing to the extreme, producing responses that bordered on aggressive sarcasm.

Phase 3: Perfect Balance

PERFECT BALANCE

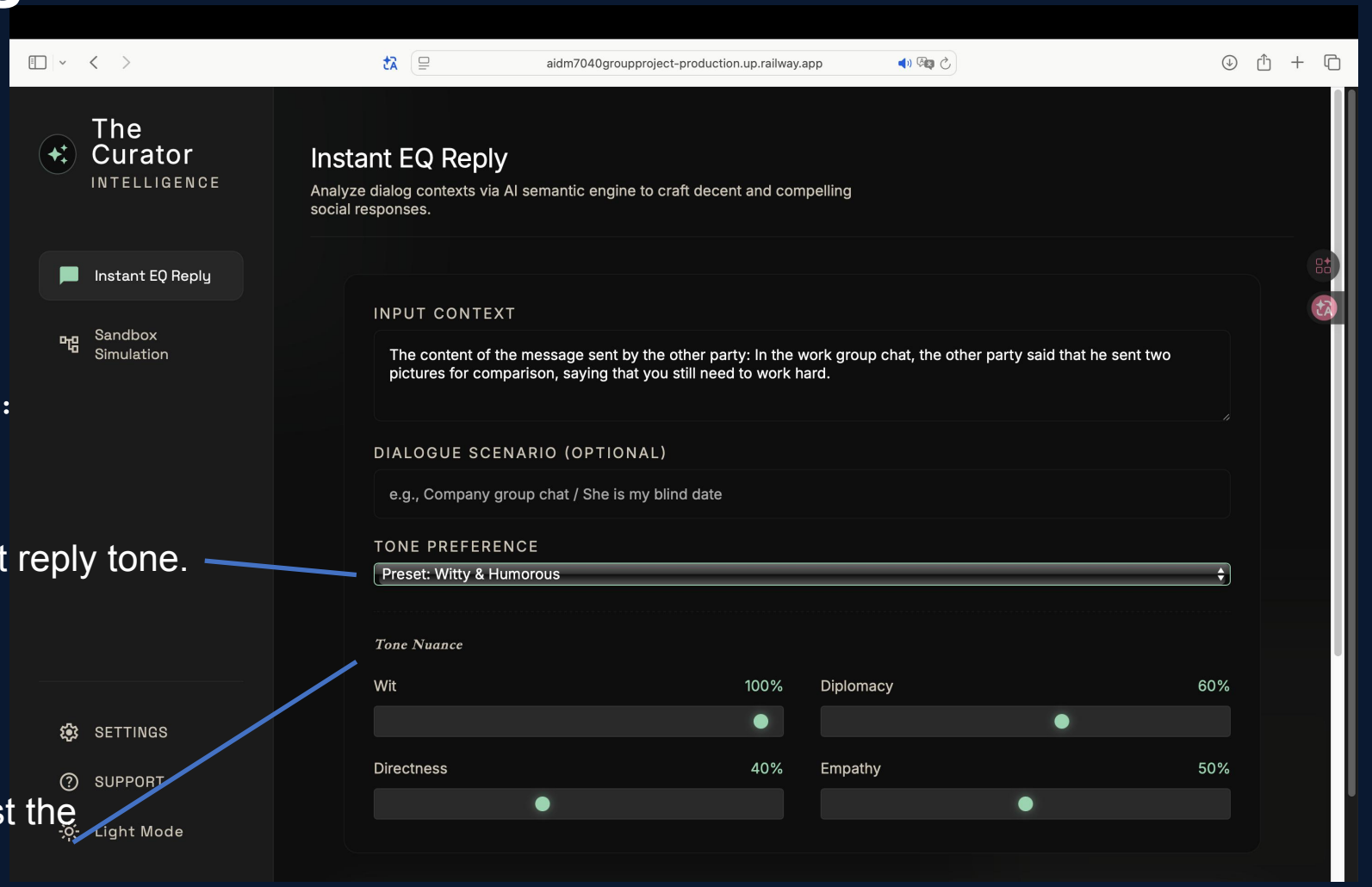
Our final, meticulously calibrated prompt framework achieved a stable, functional balance between human decency and steadfast boundary protection.

Web page architecture

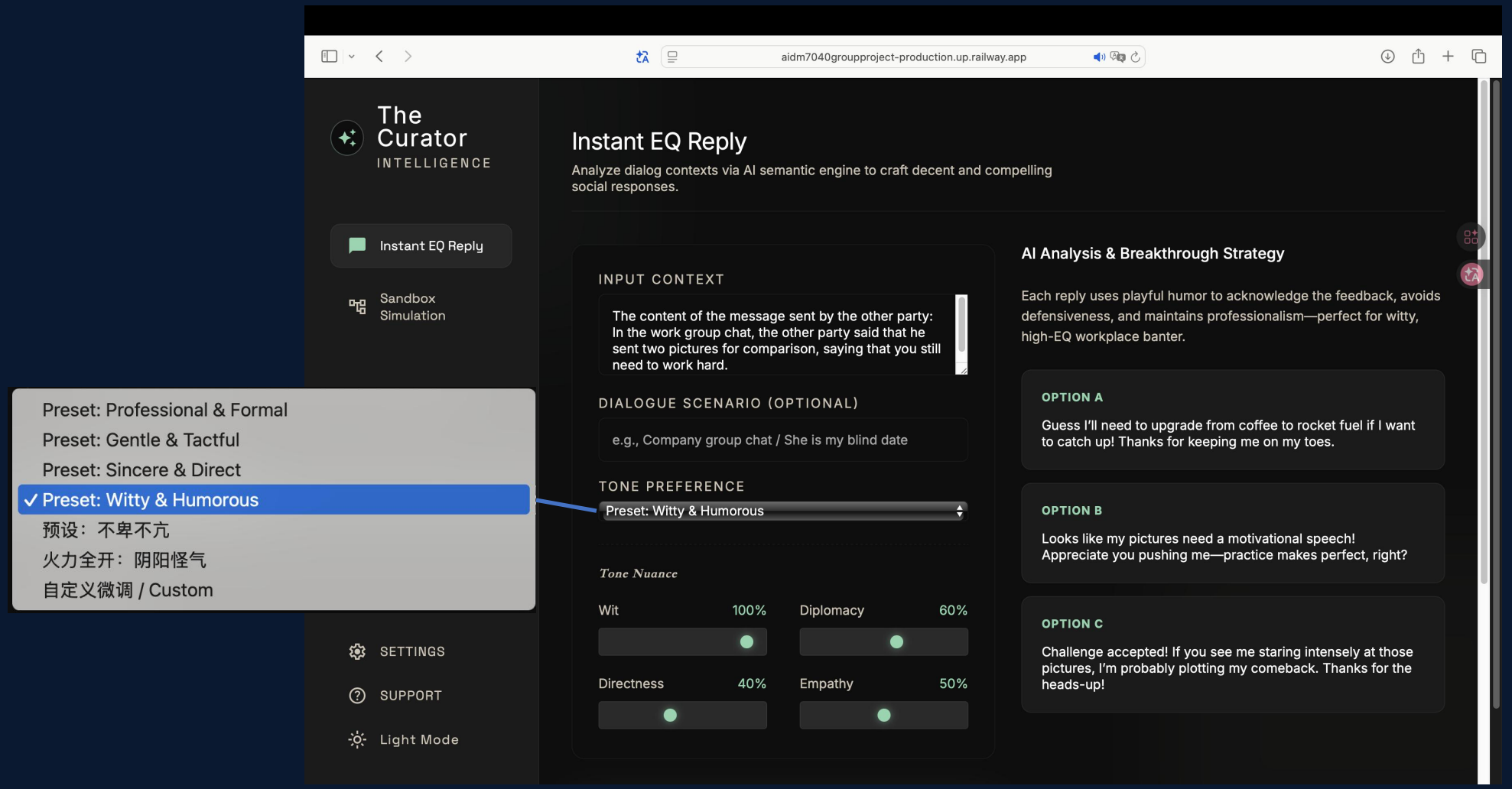
Two ways to choose the tone:

1. Choose the set reply tone.

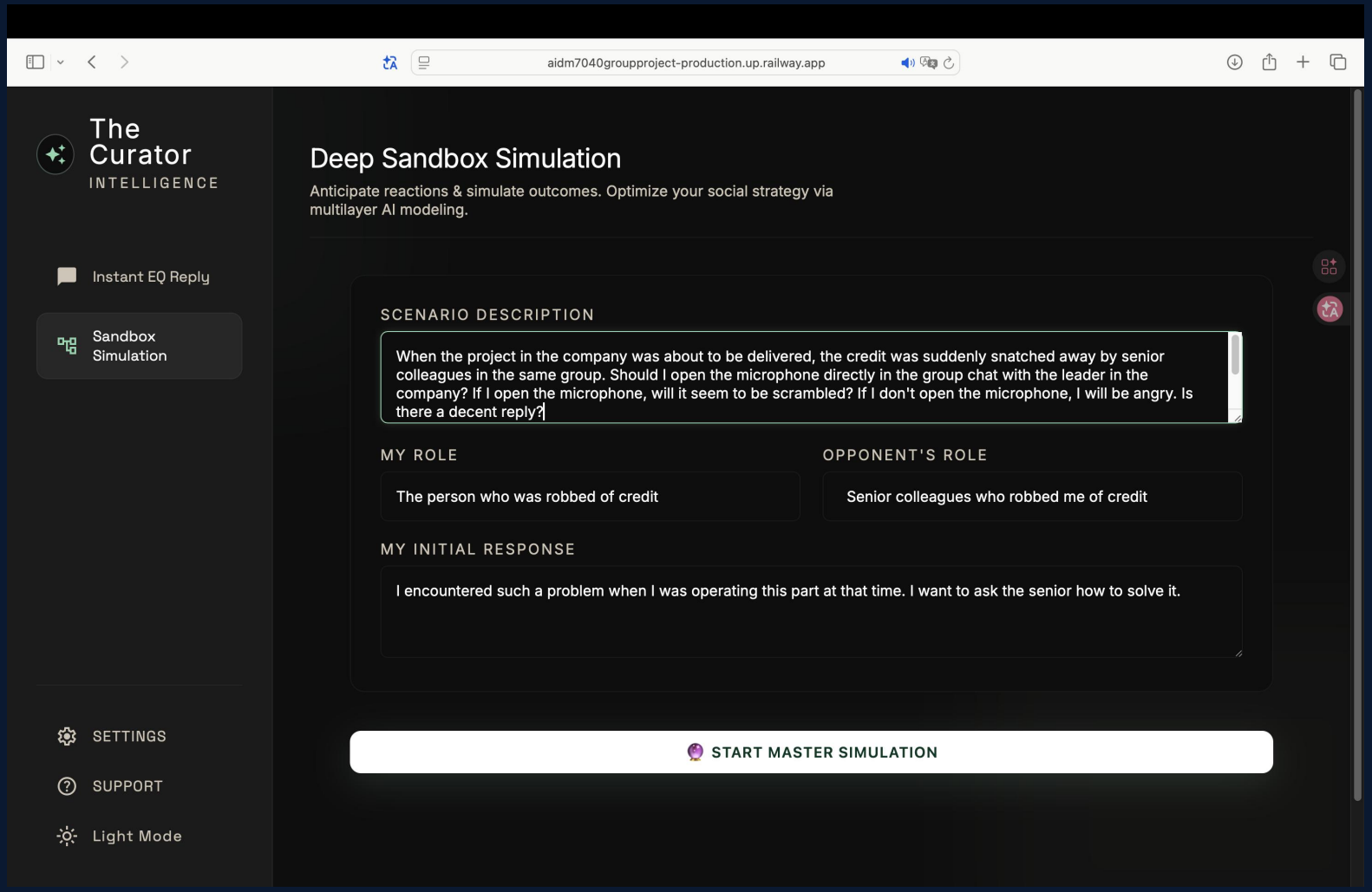
2. Manually adjust the tone composition



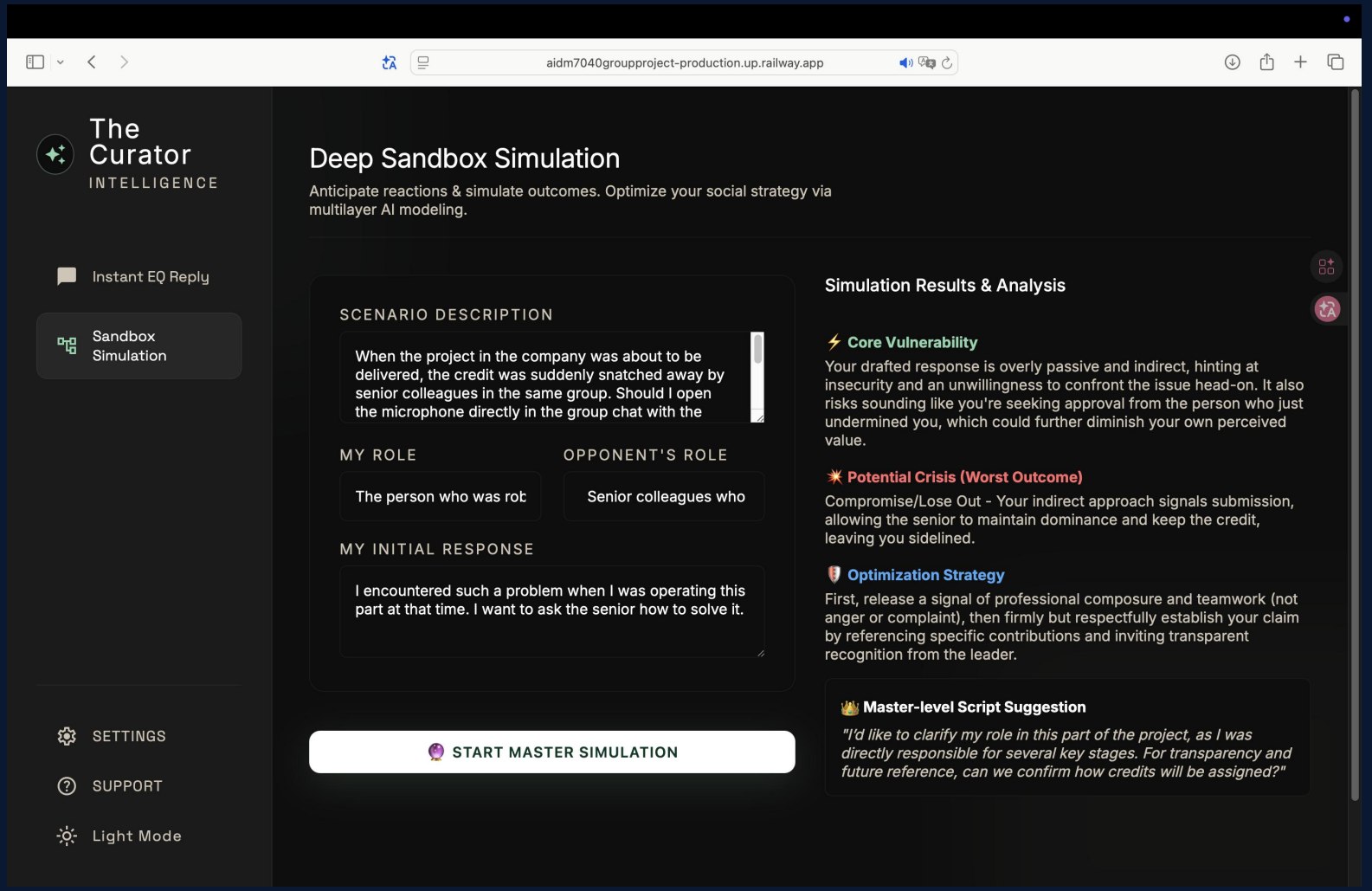
Web page architecture



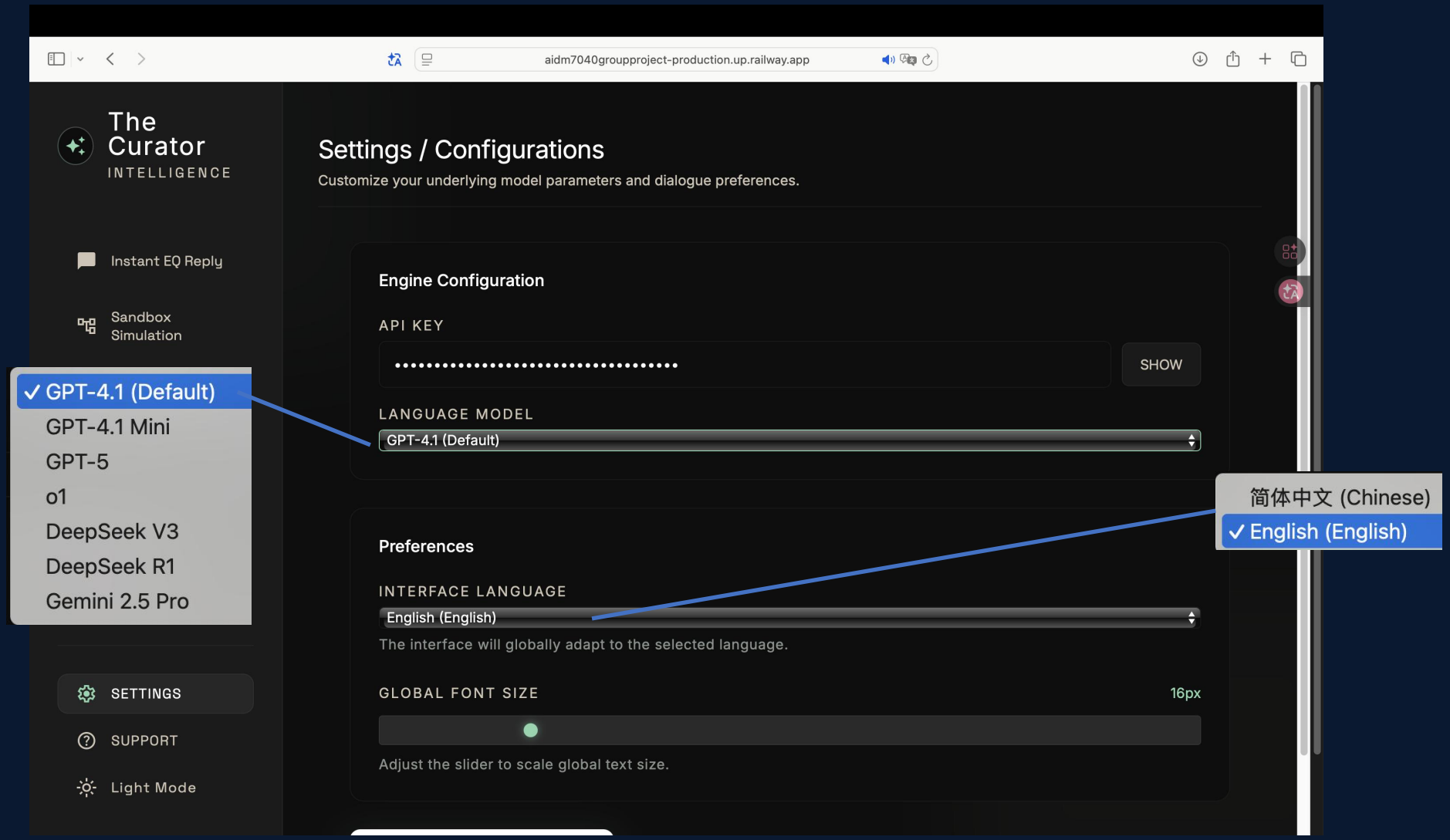
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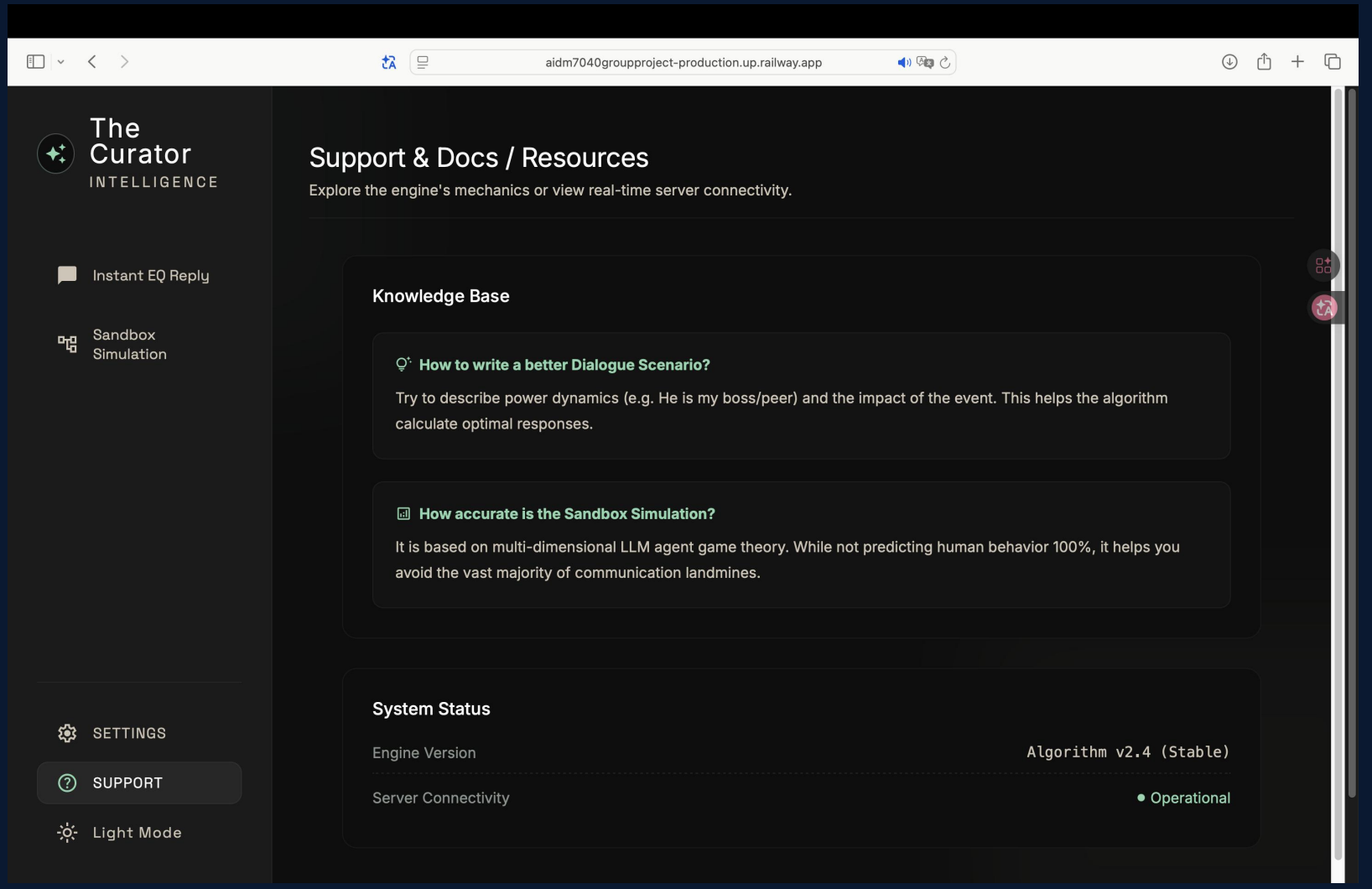
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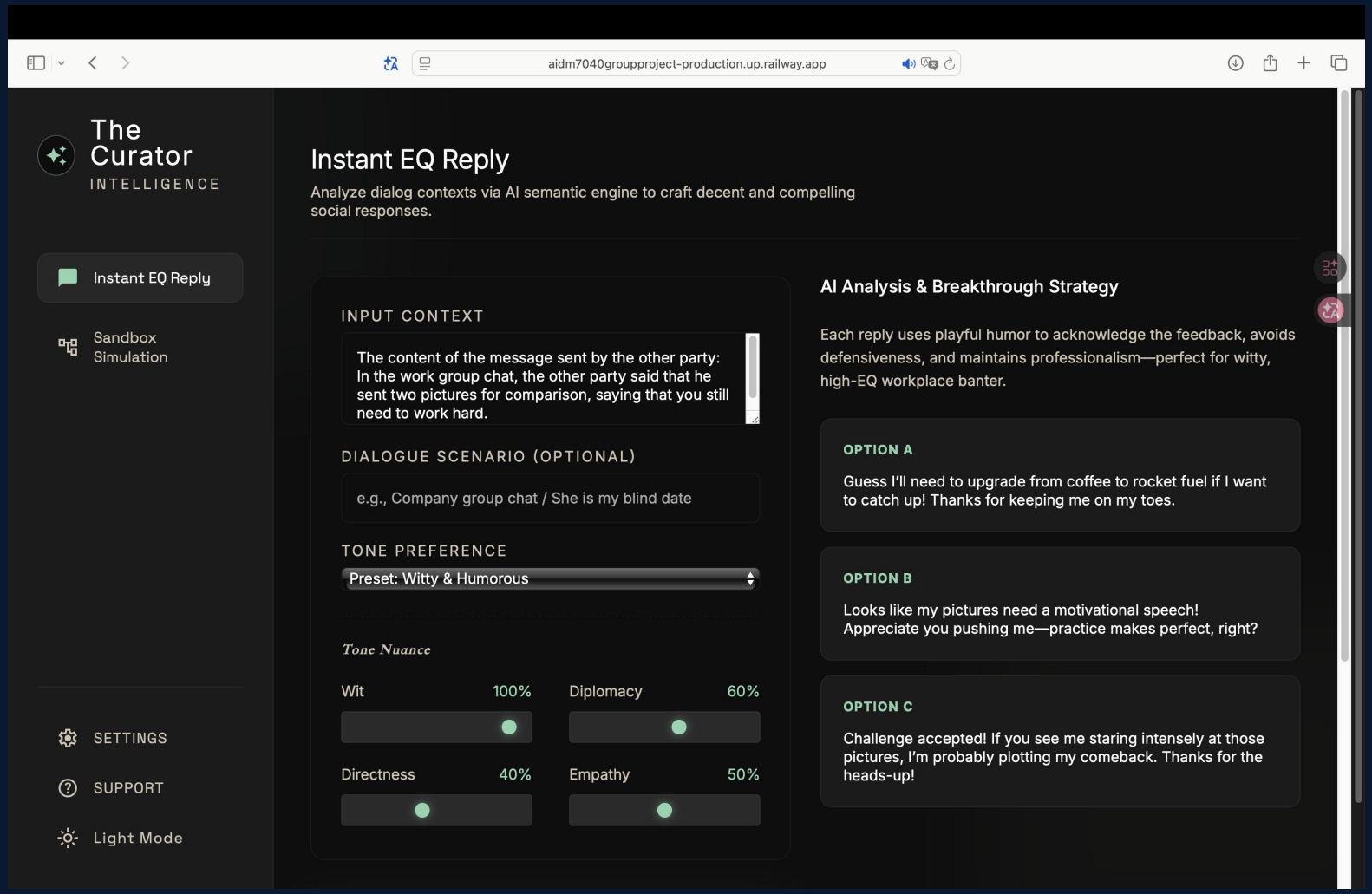


UI Design

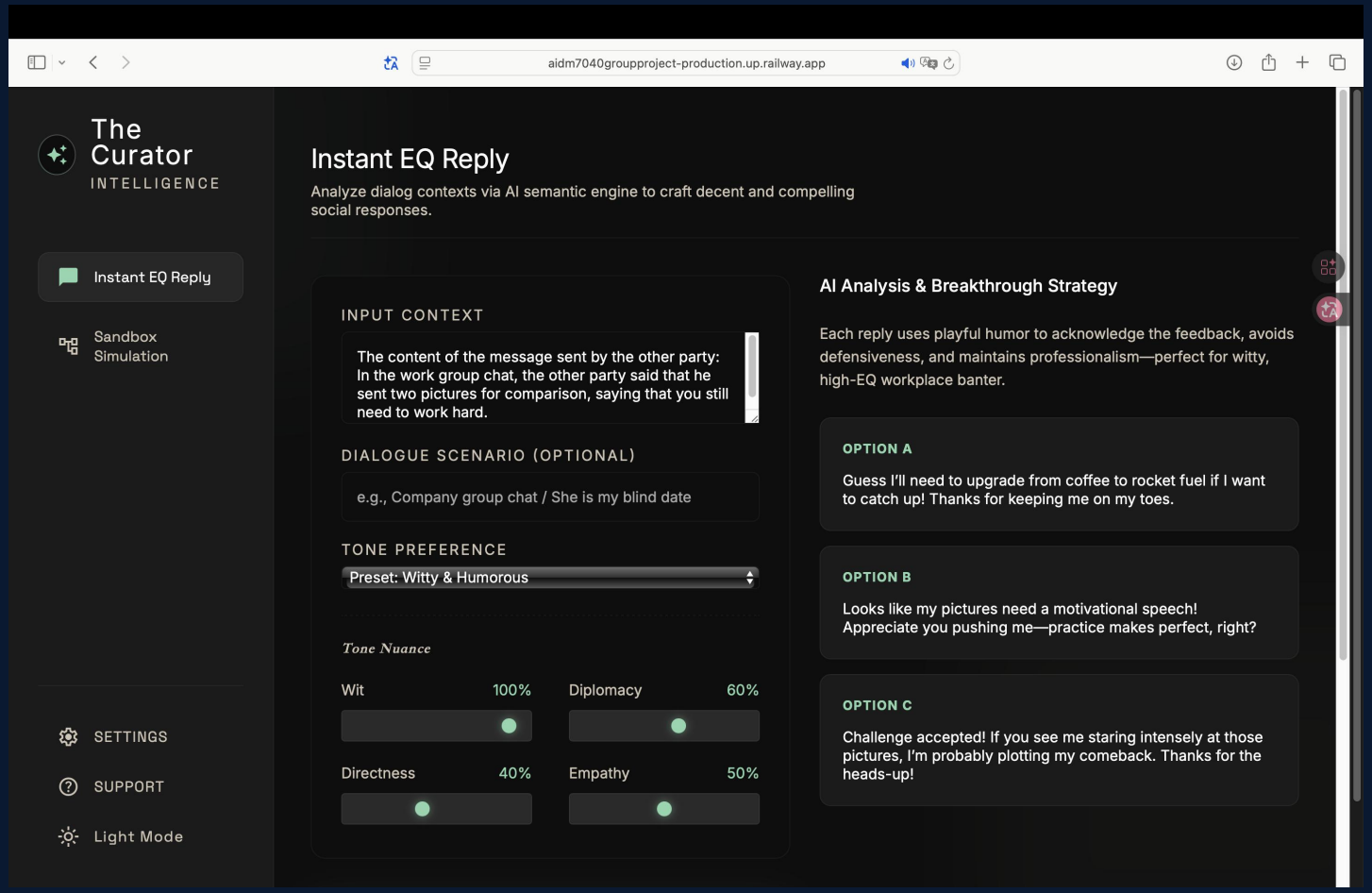
Visual Aesthetics

Features an all-black background paired with high-contrast text. This design not only aligns with the modern, high-tech aesthetic typical of AI products but also reduces visual fatigue during prolonged use—making it particularly well-suited for high-frequency usage scenarios such as professional work environments and social communication.

The use of low-contrast neutral gray text accented by highlights of bright green ensures information readability without allowing intense colors to distract the user; the overall atmosphere remains understated and unobtrusive.



UI Design



Layout Architecture

Side Navigation Bar: Employs a classic fixed-position left-side navigation bar, clearly segmented into functional areas (e.g., Instant Reply, Sandbox Simulation) and system areas (e.g., Settings, Support, Mode Switching).

Card-based Layout: The overall typography and spacing are well-balanced, avoiding the cluttered sensation of information overload and enabling users to quickly locate the specific operational areas they require.

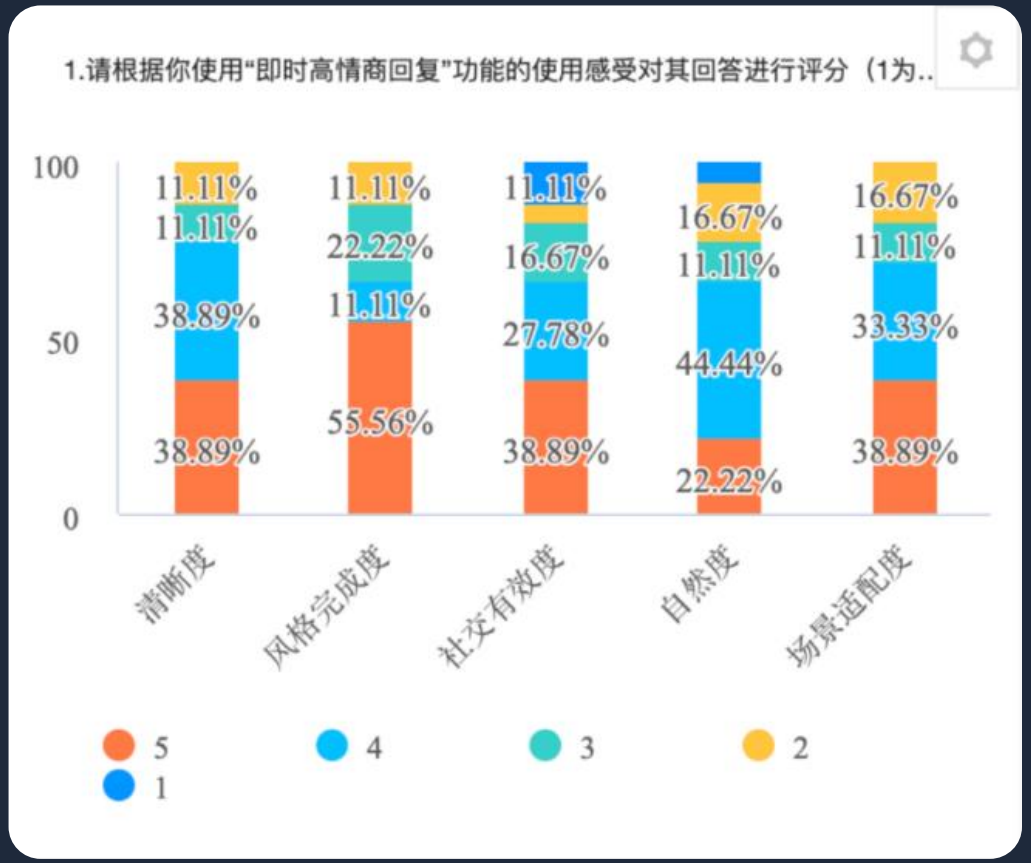
Data Validation: Blind Survey Comparison Results

According to the subsequent blind testing data, the vast majority of users preferred the fine-tuned system's output over the native LLM during the high emotional intelligence reply scenarios.

Key Breakthrough: Significant Jump in User Satisfaction

- **Boundary Management -** Establish a clear social security defense line
- **Context Appropriateness -** Accurately match the needs of complex social scenarios

The visualization of these comparative metrics illustrated a substantial aggregate increase in user satisfaction specifically within the volatile realms of Boundary Management and Context Appropriateness.



Academic & Applied Conclusion

Core Conclusion

This project demonstrates that Generative AI can be elevated from the realm of general-purpose chatbots to operate as a highly specialized, surgically precise**social survival mechanism**.

Academic Significance

Our methodology successfully neutralized the intrinsic**compliance bias**typically observed in commercial LLMs, proving that with the right algorithmic reins, these models can navigate the high-context labyrinths of human interaction.

Workload Distribution



25%

LI JIHANG

Core Technical Development

Responsible for backend programming, HKBU API integration, multi-model version mapping, and core system architecture.



25%

LI DISAI

Product Management & Prompt Engineering

Coordinates system requirements, prompt design & ablation tuning, context-aware logic, and project reporting.



25%

GUO ZILIANG

Research & Data Analytics

Designs user surveys, conducts blind tests, compiles data, builds evaluation matrices for robust project support.



25%

LI ZINING

UI/UX Design & Visualization

Handles web interface interaction & visual design, emotional quotient matrix radar chart visualization, and slide design.

Q&A

Thank You

THANKS FOR WATCHING